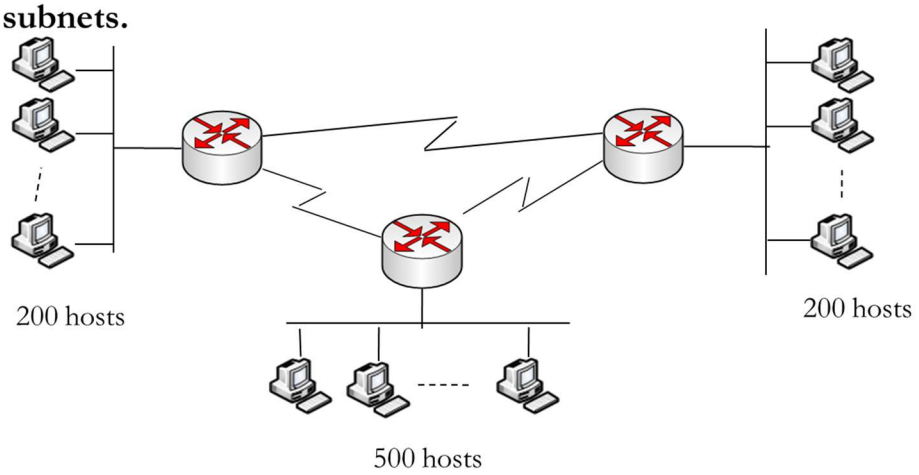
EBU5212 Class Test Two -- 2023/24

Answer All Questions below.

1. **Subnet 10.30.124.0/22 to address the network requirement below by using the most efficient addressing possible. Describe every step of your solution. List the address range for three subnets.**



[7.5 marks]

2. A (7,4) linear-block coding scheme is to be implemented with a generator matrix $\mathbf{G}$ defined as in Figure .

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Figure 1

Using a data word $\mathbf{D} = [1 \quad 0 \quad 0 \quad 1]$:

i) Calculate the codeword to be transmitted.

ii) Show that with an error in the 5th bit of the codeword the scheme can correct 1-bit errors.

iii) Show by putting the 5th and 7th bits of the codeword in error that the scheme is not resilient to 2-bit errors.

[7.5 marks]